

STAA57-Winter 2024: Worksheet-3

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Questions 1:

- i. Download the “Daily Shelter Overnight Occupancy” data from the city of Toronto’s open data portal. This following line will do that for you

<https://ckan0.cf.opendata.inter.prod-toronto.ca/dataset/21c83b32-d5a8-4106-a54f-010dbe49f6f2/resource/55d58477-50f5-490c-8da8-5953e3b26ca4/download/daily-shelter-overnight-service-occupancy-capacity-2022.csv>

- ii. Load the dataset in R studio and by using the LOCATION_CITY variable create the following summaries for the CAPACITY_ACTUAL_ROOM variable. Your output should match the dimension and order of this following output.

LOCATION_CITY	N	AVE	Quartile_1st	Quartile_3rd
Toronto	38649	55.96007	27	71
North York	4813	80.29806	14	83
Scarborough	3222	124.45297	39	131
Etobicoke	1095	146.59726	146	148
Vaughan	365	67.42466	67	68

Questions 2:

Create an animation using ganimate() where the graph shows a transition from once city to the other (use the LOCATION_CITY variable).

For each city, show the histogram of OCCUPIED_ROOMS.

You need to submit the **code only**. Do not try to create the animation as part of the pdf as it will produce many different graphs on many different pages. So use eval=FALSE option for the code chunk for this question.

Instructions for completing this worksheet:

- Create a R markdown file, use your name and student id as the author name.
- Use the command ‘\pagebreak’ between your answers to Q1 and Q2 (this will make sure when you convert to pdf, answer to Q2 starts on a new page)
- Create an html output first, open it in any browser and then print or save as a pdf. (if you already know how to create a pdf directly from R studio then disregard this step)
- Upload your pdf to crowdmark in the place designated for Q1.

- Drag the pages corresponding to Q2 to the place designated to Q2.
- Finish submitting everything by 6pm please.